DATA ENGINEERING CHEAT SHEET

A plain-English map of the terms, tools, formats, and architecture you will meet in modern data engineering.

A simple mental model: Sources -> Ingestion -> Storage -> Processing -> Modeling -> Serving / BI. Orchestration, quality, governance, and observability sit across the whole flow.

Layer	Typical examples
Storage	S3, HDFS, Parquet, Iceberg
Compute / Query	Spark, Druid, StarRocks, DuckDB, Trino
Streaming	Kafka, Flink
Orchestration	Airflow, Dagster
Transformation / Modeling	dbt, SQL, star schema
Serving / BI	Power BI, Tableau, APIs

How to use it: learn the concept first, then the product. For example, Iceberg is a table format; Druid is an OLAP database; DuckDB is an embedded analytical database; Dagster is an orchestrator.

1. Foundations & Architecture

TERM	SIMPLE DEFINITION
Data Engineering	Building systems that collect, move, clean, store, and serve data reliably.
Data Pipeline	A sequence of steps that moves and transforms data from source to destination.
ETL	Extract, Transform, Load: transform data before loading it into the target system.
ELT	Extract, Load, Transform: load raw data first, then transform it inside the analytics platform.
Batch Processing	Process a group of records together on a schedule, such as every hour or night.
Streaming	Process data continuously or nearly continuously as events arrive.
Real-time / Near real-time	Data is available seconds or minutes after it is created, rather than hours later.
Source System	The system where data originates, such as an app database, SaaS tool, logs, or sensors.
Sink / Destination	The system receiving data, such as a warehouse, lake, search engine, or application.
Data Platform	The combined infrastructure and tools used to ingest, store, process, govern, and query data.

2. Storage: Lake, Warehouse & Lakehouse

TERM	SIMPLE DEFINITION
Data Lake	Low-cost storage for large amounts of raw or processed data, commonly on object storage.
Data Warehouse	Analytics database optimized for structured data, SQL, reporting, and BI workloads.
Lakehouse	Architecture that adds warehouse-like tables, transactions, and governance to a data lake.
Data Mart	A smaller analytics store focused on one team or business area, such as finance or sales.
Object Storage	Highly scalable file/object storage such as Amazon S3, Azure Blob, or Google Cloud Storage.
HDFS	Hadoop Distributed File System; stores large files across multiple machines with replication.
Hot / Warm / Cold Storage	Storage tiers trading cost for speed: hot is fastest, cold is cheapest and slower.

3. File & Table Formats

TERM	SIMPLE DEFINITION
CSV	Simple text format where values are separated by commas; easy to use but not optimized for analytics.
JSON	Human-readable nested data format common in APIs, logs, and event streams.
Avro	Row-oriented binary format with a schema; useful for events, serialization, and streaming.
Parquet	Columnar file format optimized for analytics, compression, and reading only needed columns.
ORC	Columnar analytics format similar to Parquet, widely used in the Hadoop ecosystem.
Open File Format	A publicly specified format readable by many tools, helping avoid dependence on one vendor.
Apache Iceberg	Open table format for large analytic tables on object storage, with snapshots and safe schema changes.
Delta Lake	Open table format adding ACID transactions, versions, and metadata to files in a data lake.

TERM	SIMPLE DEFINITION
Apache Hudi	Open table format focused on incremental updates, upserts, and change-heavy lake workloads.
Manifest / Metadata File	Files that describe which data files belong to a table or snapshot so engines can plan reads efficiently.

4. Databases & Analytics Engines

TERM	SIMPLE DEFINITION
OLTP	Online Transaction Processing: databases optimized for many small inserts, updates, and point lookups.
OLAP	Online Analytical Processing: systems optimized for scans, aggregations, joins, and reporting.
Row Store	Stores values from each row together; generally good for transactional access.
Column Store	Stores values by column; usually better for analytical scans and compression.
MPP	Massively Parallel Processing: splits a query across many machines or CPU cores.
Apache Druid	Distributed real-time OLAP database designed for fast slice-and-dice analytics on event/time-series data.
StarRocks	MPP analytical database built for fast SQL analytics, including high-concurrency and real-time use cases.
ClickHouse	Column-oriented analytical database known for very fast aggregation and event/log analytics.
DuckDB	Embedded analytical SQL database that runs inside a process and can query local files such as Parquet directly.
Trino	Distributed SQL query engine that queries many different data sources without requiring all data to be copied first.
Presto	Distributed SQL engine originally created at Facebook; an ancestor/sibling ecosystem to Trino.
Apache Spark SQL	SQL engine within Spark for distributed processing of structured data.
Snowflake	Cloud data warehouse/platform that separates storage from compute and manages infrastructure for you.
BigQuery	Google Cloud's serverless analytics warehouse for large-scale SQL queries.
Amazon Redshift	AWS cloud data warehouse optimized for analytical SQL workloads.

5. Hadoop & Distributed Processing

TERM	SIMPLE DEFINITION
Apache Hadoop	Ecosystem for distributed storage and batch processing; historically centered on HDFS and MapReduce.
MapReduce	Programming model that processes data using map and reduce stages across a cluster.
YARN	Hadoop resource manager that allocates cluster CPU and memory to applications.
Apache Spark	Distributed compute engine for batch, SQL, streaming, and machine-learning workloads.
RDD	Spark's low-level resilient distributed collection abstraction.
DataFrame	Table-like distributed data abstraction with named columns and query optimizations.
Shuffle	Redistributing data between workers, often required for joins or group-bys; usually expensive.
Partition	A logical split of data or work so different pieces can be stored or processed independently.

TERM	SIMPLE DEFINITION
Cluster	A group of machines working together as one data processing or storage system.
Worker / Executor	A process or machine that performs assigned computation in a distributed job.

6. Streaming & Messaging

TERM	SIMPLE DEFINITION
Event	A record describing something that happened, such as an order created or button clicked.
Event Stream	An ordered or semi-ordered continuous flow of events.
Apache Kafka	Distributed event streaming platform used to publish, store, and consume streams of records.
Topic	Named stream/category in a messaging system such as Kafka.
Kafka Partition	Ordered shard of a Kafka topic that enables parallelism and scaling.
Producer	Application that writes messages/events to a broker or stream.
Consumer	Application that reads messages/events from a broker or stream.
Consumer Group	Set of consumers that share the work of reading partitions from a topic.
Offset	Position of a message inside a Kafka partition; used to track consumption progress.
Apache Flink	Distributed stream-processing engine designed for stateful, event-time processing.
Spark Structured Streaming	Spark API for treating streams as continuously updated tables/dataframes.
Watermark	A rule for deciding how late an event may arrive before a streaming window is considered complete.
Exactly-once	Processing guarantee aiming for each logical event to affect results only once despite retries/failures.
At-least-once	Delivery guarantee where events are not lost but may be processed more than once.

7. Ingestion, CDC & Integration

TERM	SIMPLE DEFINITION
Data Ingestion	Bringing data from source systems into the data platform.
CDC	Change Data Capture: capture inserts, updates, and deletes from a source database as changes happen.
Log-based CDC	CDC that reads the database transaction log instead of repeatedly querying tables.
Debezium	Open-source CDC platform that turns database changes into event streams, commonly through Kafka.
Kafka Connect	Framework for standardized source and sink connectors around Kafka.
Airbyte	Data integration platform providing connectors for moving data between many sources and destinations.
Fivetran	Managed ELT service that extracts data from SaaS/databases into analytical destinations.
Reverse ETL	Move modeled warehouse data back into operational tools such as CRM or marketing platforms.
Backfill	Reprocess historical data to fill missing periods or apply new logic to old records.

8. Orchestration & Transformation

TERM	SIMPLE DEFINITION
Orchestration	Scheduling jobs, managing dependencies, retries, parameters, and observability across pipelines.
DAG	Directed Acyclic Graph: tasks connected by dependencies with no circular path.
Apache Airflow	Workflow orchestrator where pipelines are commonly defined as Python DAGs.
Dagster	Data orchestrator centered on assets, dependencies, testing, lineage, and software-defined data workflows.
Prefect	Python workflow orchestration platform designed for flexible execution and monitoring.
dbt	Tool for transforming data with SQL, tests, documentation, and dependency graphs inside analytical databases.
Data Asset	A durable data product such as a table, file, model, or dataset produced by a pipeline.
Idempotency	Property where safely repeating an operation produces the same final result as doing it once.
Retry	Automatically run a failed task again, usually with limits and delays.
Checkpoint	Saved progress/state that lets a processing job resume without starting from the beginning.

9. Modeling & Warehouse Design

TERM	SIMPLE DEFINITION
Data Model	Structured representation of entities, relationships, measures, and dimensions used for analysis.
Fact Table	Table containing measurable business events such as orders, clicks, or payments.
Dimension Table	Descriptive table used to categorize facts, such as customer, product, date, or region.
Star Schema	Warehouse model with a central fact table connected directly to dimension tables.
Snowflake Schema	More normalized version of a star schema where dimensions are split into related tables.
Surrogate Key	Generated identifier used instead of a source-system business key.
SCD	Slowly Changing Dimension: techniques for handling changes to dimension attributes over time.
Denormalization	Duplicate or combine data to reduce joins and improve read performance.
Normalization	Organize data into related tables to reduce duplication and update anomalies.
Semantic Layer	Business-friendly definitions for metrics and dimensions shared consistently across analytics tools.

10. Query & Performance Terms

TERM	SIMPLE DEFINITION
Predicate Pushdown	Apply filters as close to the storage layer as possible so less data is read.
Partition Pruning	Skip partitions that cannot match the query filter.
Column Pruning	Read only columns required by the query.
Index	Extra data structure that helps locate matching records faster.
Materialized View	Precomputed query result stored so repeated analytical queries can run faster.
Query Plan	The execution strategy chosen by the database for scans, joins, filters, aggregations, and exchanges.
Cost-Based Optimizer	Optimizer that estimates alternative plans and chooses one expected to be cheapest.
Hash Join	Join technique that builds a hash table from one input and probes it with the other.

TERM	SIMPLE DEFINITION
Broadcast Join	Send a small table to all workers so a large distributed table does not need to move.
Compaction	Merge many small data files or storage fragments into fewer/larger ones for better efficiency.
Small Files Problem	Too many tiny files cause excessive metadata work and poor scan performance.

11. Reliability, Quality & Governance

TERM	SIMPLE DEFINITION
Data Quality	How correct, complete, timely, consistent, and valid data is for its intended use.
Data Validation	Checks that data matches expected rules, types, ranges, formats, and relationships.
Schema	Definition of the fields/columns, types, and structure of a dataset.
Schema Evolution	Changing a schema over time while keeping old and new data usable.
Data Contract	Agreement defining the structure, meaning, quality, and ownership expectations for shared data.
Data Lineage	Record of where data came from, what transformed it, and where it is used.
Data Catalog	Searchable inventory of datasets, schemas, ownership, descriptions, and metadata.
Metadata	Data about data, such as schema, location, timestamps, owners, statistics, and lineage.
Data Governance	Policies and processes controlling data ownership, quality, security, access, and lifecycle.
PII	Personally Identifiable Information: data that can identify or be linked to a person.
RBAC	Role-Based Access Control: permissions are granted to roles, then users receive roles.
Observability	Ability to understand pipeline and data health through metrics, logs, traces, lineage, and checks.
SLA / SLO	Target expectations for service or pipeline behavior, such as freshness, availability, or completion time.

12. Cloud & Infrastructure

TERM	SIMPLE DEFINITION
Compute	CPU and memory resources used to execute data processing or queries.
Storage-Compute Separation	Keep persistent data storage independent from query/processing compute so each can scale separately.
Serverless	Cloud model where the provider manages servers and capacity; users mainly submit workloads and pay for usage.
Container	Packaged application plus dependencies that runs consistently across environments.
Docker	Popular platform for building and running containers.
Kubernetes	System for deploying, scaling, networking, and recovering containerized applications across clusters.
Autoscaling	Automatically add or remove compute resources based on workload.
IAM	Identity and Access Management: controls who or what can access cloud resources.
VPC	Virtual Private Cloud: logically isolated network environment in a cloud provider.

13. BI, Analytics & Serving

TERM	SIMPLE DEFINITION
BI	Business Intelligence: dashboards, reports, and analysis used to support business decisions.
Power BI	Microsoft analytics and visualization platform for reports, semantic models, and dashboards.
Tableau	Business intelligence platform for interactive data visualization and dashboards.
Looker	Google Cloud BI platform with a governed modeling layer and dashboards.
DirectQuery	BI mode where visualizations query the source database live instead of importing all data into the BI tool.
Import Mode	BI mode where data is copied into the BI tool's own in-memory/model storage for faster interaction.
Metric	A numeric business measurement such as revenue, active users, or conversion rate.
Dimension	An attribute used to group or filter metrics, such as country, product, or date.
Serving Layer	System optimized to expose curated data to BI, APIs, applications, or downstream consumers.

14. Modern Interoperability

TERM	SIMPLE DEFINITION
Apache Arrow	In-memory columnar data format designed for fast data interchange between systems.
Arrow Flight	High-performance RPC protocol for transferring Arrow data over the network.
Arrow Flight SQL	SQL-oriented protocol built on Arrow Flight for querying and transferring tabular results efficiently.
JDBC	Java standard API/driver model for connecting applications to relational databases.
ODBC	Language-neutral database connectivity standard widely used by BI and desktop tools.
Connector	Software adapter that lets one system read from or write to another.
Catalog	In query engines, a configured namespace/connection that exposes databases or external data sources.
Federated Query	Query data across multiple independent systems through one query engine or interface.

Rule of thumb: when you hear a new data term, ask: Is it a storage system, file/table format, compute/query engine, messaging system, orchestrator, or BI/serving tool? That classification removes most of the confusion.